

BF Lab - User Manual

WatchOS on Waveshare ESP32-S3 Touch AMOLED 2.06

App: BF Lab (id: brainfuck) | Version 1.2.0 | 2026-09-18

1. What it is

BF Lab is an on-watch Brainfuck editor and runner. Write programs with the eight classic ops, save them on the SD card, run step-by-step or continuously, and optionally drive GPIO pins or draw pixels through special tape cells. No QWERTY - only the BF keypad.

2. How to open it

- 1) Unlock the watch (clock/PIN).
- 2) Open the app launcher (icon grid by default, or list in Settings).
- 3) Tap BF Lab.
- 4) Home buttons: Edit, Files, Run, Map, Ex, Exit.
- 5) System Back / left swipe moves toward Home, then exits.

3. Screens and controls

Mode	What you do
-----	-----
Home	Edit / Files / Run / Map / Ex / Exit
Edit	Keypad > < + - . , [] plus BS / CLR. Cursor shown as
Files	/sd/bf/*.bf - New, Save, Load, Del, Prev, Next
Run	Go, Stop, Step, Reset; tape; draw canvas; . out; , in
Map	GPIO + pixel register map; Pulse0 / Low0 / ClrDraw
Ex	Demos: ASCII A, GPIO60, Plot, FB chk, loop -> Home

Edit tips: max 192 ops; non-BF chars stripped on load; BS deletes left of cursor; CLR clears program.

Run tips: Go runs up to 32 steps per UI tick; Step = one op; Stop halts; Reset rebuilds brackets. '.' appends printable ASCII (else \HH). ',' reads Run input (empty=0). In+A queues A (65); InClr clears input.

4. Saving programs (SD)

```
/sd/bf/prog001.bf
/sd/bf/prog002.bf
...
```

Insert the microSD card. If the PC has the card via USB mass storage, eject first. First Files/Save creates /sd/bf/. Files are plain-text .bf (ops only).

5. Brainfuck quick reference

Op	Meaning
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>	Move pointer right (wraps on 64-cell tape)
<	Move pointer left
+	Increment cell (0..255 wrap)
-	Decrement cell
.	Output cell as ASCII

```
,      Input into cell
[      Jump past ] if cell is 0
]      Jump back to [ if cell is not 0
```

Tape: 64 cells (0-63). Cells 0-47 are ordinary RAM for classic programs.

6. GPIO tape map

Tape cell	GPIO pin
-----	-----
60	10
61	16
62	19
63	20

Values 0/1 = digital; values >1 attempt PWM. Ex->GPIO60 pulses pin 10. Map screen has Pulse0/Low0 helpers.

7. Pixel I/O (v1.2)

Pure BF - no new symbols. Mid-tape cells are graphics registers. A 16x16 framebuffer is shown as a 64x64 canvas on Run (each FB pixel = 4x4). Firmware includes watch.canvas_pixel with BF Lab 1.2.

Cell	Role
-----	-----
48	Mode: 0=normal, 1=FB address mode (X/Y are 0..15)
49	FB cursor (linear 0..255); auto-updated on poke
50	X - canvas 0..63 (plot) or FB 0..15 (mode 1)
51	Y - same
52	R 0..255
53	G 0..255
54	B 0..255
55	Plot strobe - write nonzero to paint; auto-clears
56	Brush size 1..8 (default 1)
57	Clear-draw strobe - canvas_clear; auto-clears
58	FB poke - write RGB into FB; paints 4x4; auto-clears

Tip: on a zero cell, '-' wraps to 255 (bright RGB). Classic programs using only cells 0-47 are unaffected. GPIO 60-63 still work while drawing.

8. Built-in examples (Ex)

Name	What it shows
-----	-----
ASCII A	Output letter A
GPIO60	Pulse GPIO 10 via tape[60]
Plot	RGB/XY + strobe cell 55
FB chk	Framebuffer checker demo
loop	Simple loop

Ex loads the program and returns to Home. Open Run -> Go to execute.

9. Limits

- Editor max: 192 ops
- Tape: 64 cells, 8-bit wrap
- Run burst: 32 steps/tick
- SD required for Files/Save; USB MSC lock blocks SD - eject first
- Only GPIO 10/16/19/20 from BF
- Draw path avoids many LVGL rects (stability)

10. Quick start

- 1) Launcher -> BF Lab
- 2) Ex -> Plot (or FB chk) -> Run -> Go
- 3) Edit a tiny program -> Files -> Save -> Run -> Go
- 4) Use Map while learning registers

11. Troubleshooting

Symptom	Try
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No SD / empty list	Insert card; eject USB storage; retry Files
GPIO silent	Cells 60-63 only; check pins 10/16/19/20
No pixels	Ex->Plot; need canvas_pixel (BF 1.2 flash)
Slow long program	Normal (32 steps/tick); use Step
Program too long	Stay under 192 ops or split files

Reference: docs/BRAINFUCK.md | data/apps/brainfuck/ | WatchOS agent